MySQL Practice - Scenario-Based Questions

Question 1 - Library Management

A library wants to maintain book details in MySQL.

1. Create a database named **library_db**.
2. Use the **library_db** database.
3. Create a **book** table with:

   book_id, book_name, author, category, price
4. Insert details of **5 books**.
5. Display all book details.
6. Add a **publisher** column.
7. Add the **publisher** column after **book_name**.
8. Modify **book_name** to **VARCHAR(100)**.
9. Change the column **price** to **book_price**.
10. Update the **category** of a book using its **book_id**.
11. Delete one book using its **book_id**.
12. Display books where **book_price > 500**.
13. Display books where **book_price > 500** AND category is 'Programming'.
14. Display books ordered by **book_price** from highest to lowest.

Question 2 - Restaurant Management

A restaurant wants to store its menu items.

1. Create a database named **restaurant_db**.
2. Use the database.
3. Create a **menu** table with:

   item_id, item_name, category, price, availability
4. Insert details of **5 menu items**.
5. Display all menu items.
6. Add a **rating** column after **item_name**.
7. Modify **item_name** to **VARCHAR(100)**.
8. Change **price** to **item_price**.
9. Update the **availability** of a menu item using its **item_id**.

10. Delete one menu item using **item_id**.
11. Display items where **item_price > 200**.
12. Display items where **item_price > 200** AND availability = 'Yes'.
13. Display all items ordered by **item_price** from highest to lowest.

Question 3 - Employee Management

A company wants to maintain employee information.

1. Create a database named **company_db**.
2. Use the database.
3. Create an **employee** table with:

   employee_id, employee_name, department, salary, experience
4. Insert details of **5 employees**.
5. Display all employees.
6. Add **email** and **phone** columns after **employee_name**.
7. Modify **employee_name** to **VARCHAR(100)**.
8. Change **salary** to **monthly_salary**.
9. Update the **department** of an employee using **employee_id**.
10. Delete one employee using **employee_id**.
11. Display employees whose **monthly_salary > 30000**.
12. Display employees whose **monthly_salary > 30000** AND experience > 2.
13. Display employees ordered by **monthly_salary** from highest to lowest.

Question 4 - Movie Ticket Booking

A cinema wants to store movie information.

1. Create a database named **cinema_db**.
2. Use the database.
3. Create a **movie** table with:

   movie_id, movie_name, language, ticket_price, available_seats
4. Insert details of **5 movies**.
5. Display all movie details.
6. Add **genre** and **rating** columns after **movie_name**.
7. Modify **movie_name** to **VARCHAR(100)**.
8. Change **ticket_price** to **price**.
9. Update the **language** of a movie using **movie_id**.

10. Delete one movie using **movie_id**.
11. Display movies where **price > 200**.
12. Display movies where **price > 200** AND available_seats > 20.
13. Display movies ordered by **price** from highest to lowest.

Question 5 - Online Course Management

An online learning platform wants to store course enrollment details.

1. Create a database named **course_db**.
2. Use the database.
3. Create an **enrollment** table with:

   enrollment_id, student_name, course, fee, duration
4. Insert details of **5 students**.
5. Display all enrollment details.
6. Add **email** and **phone** columns after **student_name**.
7. Modify **course** to **VARCHAR(100)**.
8. Change **fee** to **course_fee**.
9. Update the **course name** of a student using **enrollment_id**.
10. Delete one enrollment using **enrollment_id**.
11. Display students whose **course_fee > 10000**.
12. Display students whose **course_fee > 10000** AND duration > 3.
13. Display all enrollments ordered by **course_fee** from highest to lowest.